AUTOMATED SCALE: VENDING AI-READY LZS

Integrating the AI Landing Zone Framework to Scale Enterprise AI

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About The Presenter

- **Microsoft MVP:** Recognized expert in Azure Core Technologies.
- **Cloud Architect:** Senior consultant specializing in enterprise scaling & governance.
- **IaC Advocate:** Deeply focused on automated landing zones, governance pipelines, and enterprise scale architectures.

With Sincere Thanks

"A massive thank you to the organizers, sponsors, and community members. Events like this are the bedrock of active technical evolution."

Azure Landing Zone Foundations

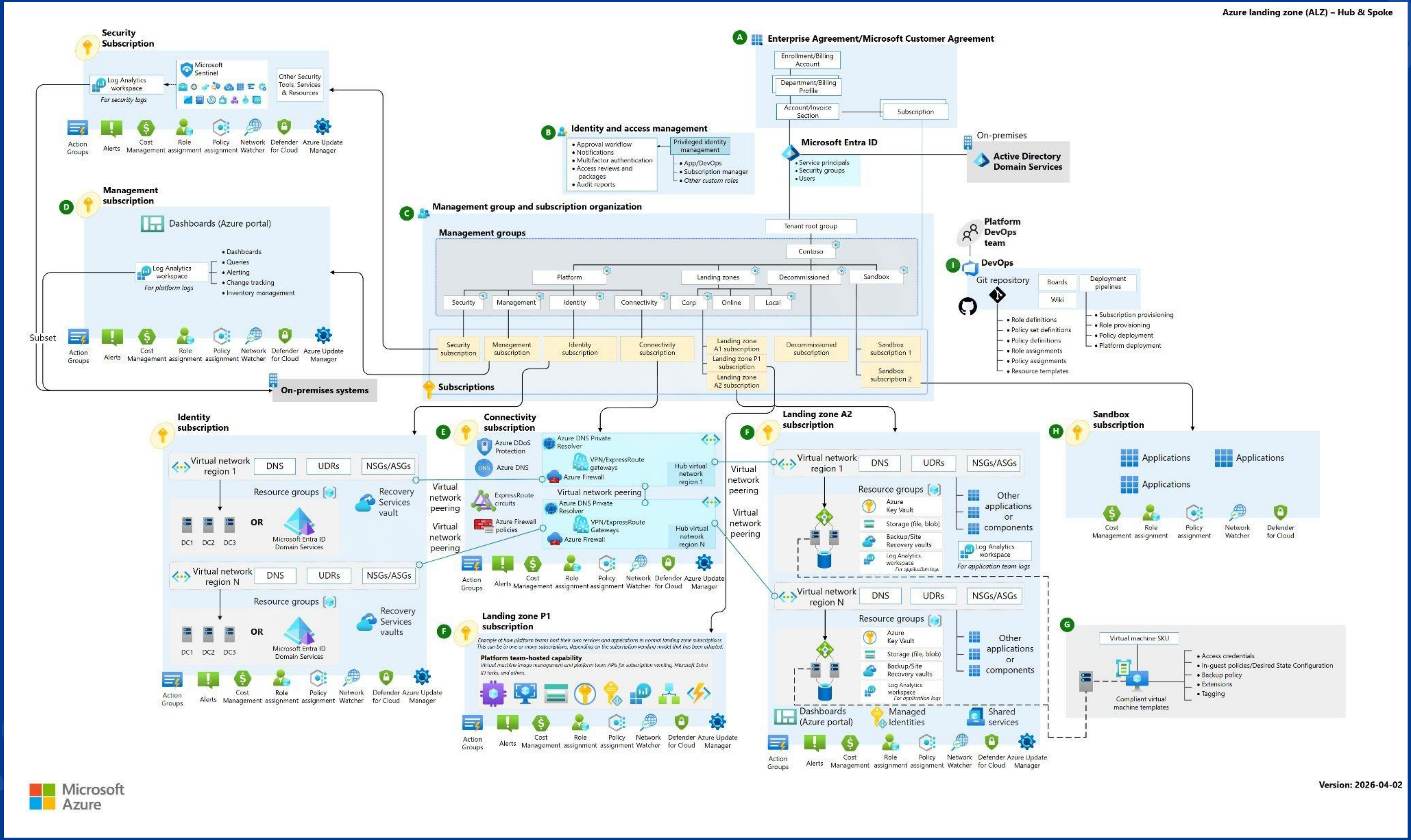
Enterprise "Scaffolding"

ALZs provide the critical foundation for cloud architecture. They ensure that Identity, Security, Topology, and Governance guidelines are programmatically established from day one across the entire Azure estate.

Microsoft CAF Blueprints

Aligned tightly with the Microsoft Cloud Adoption Framework. We leverage battle-tested subscription structures to maintain consistent policies, centralized management groups, and global standards.

Azure Landing Zone Foundations



Platform vs. Application LZs

Platform LZs (The Hub)

Dedicated to the core infrastructure owned by Central IT / CCoE teams.

- Centralized networking (Hub VNets, ExpressRoute).
- Key Vaults, Identity Management, and Logging.
- Centralized firewall and gateway routing systems.

Application LZs (The Spoke)

Isolated, workload-specific environments provisioned for application teams.

- Developer autonomy within defined secure limits.
- Direct subscription vending mapped to business units.
- Automatic enforcement of guardrails on creation.

Anatomy of anApplication LZ

Identity

Role-Based Access Control (RBAC) maps cleanly to Active Directory groups, ensuring zero wildcard privileges.

Connectivity

Dynamic VNet peering with the hub network, integrating custom DNS and restricted outbound paths securely.

Azure Policy

Strict resource-vending policies block unauthorized region deployments or public endpoint creation.

Identifying YourBaseline

100%

Secure-by-Design Compliance

The CCoE Baseline Objective

Define what your workloads require before the first application service goes online.

- **Zero Blind Spots:**Stop hoping developers will configure resources safely; build constraints directly into the vended foundation.
- **Maturity Assessment:**Standardize identity, logging, and egress rules statically across all business departments.

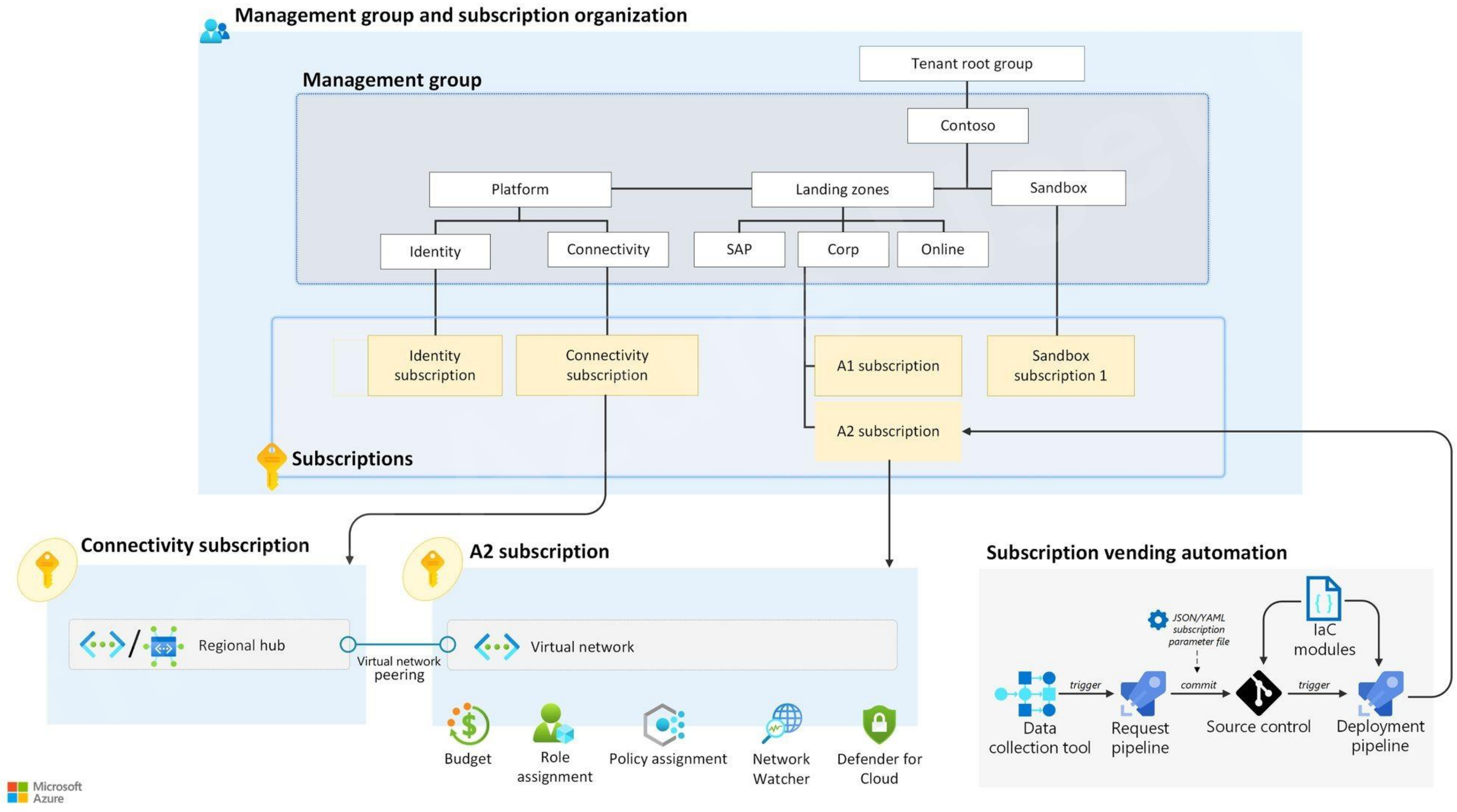
Vending: The Delivery Vehicle

Vending Over Manual Provisioning

Manual execution blocks speed-to-market and inevitably results in misconfigurations. Automated Subscription Vending injects compliance specifications directly into a subscription's DNA via Infrastructure-as-Code.

- **Minutes, Not Weeks:** Provision robust, peered, and protected subscriptions instantly.
- **Empowered Teams:** Give development pipelines direct access to sandbox environments that strictly cannot leak data.
- **Continuous Guardrails:** Automatic linking of resource groups to designated centralized tracking structures.

Vending: Microsoft's Pattern



What is anAI Landing Zone?

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An AI Landing Zone is not a totally separate cloud architecture; it is an accelerated configuration profile mapped inside your current secure CAF framework.

"

— Transforming generic application spokes into secure AI platforms.

The AI Layering Strategy

What specific infrastructure layers are added to the standard vended spoke during creation?

1. Azure Open AI

Laying down an an Azure AI resource ready for consumption on Day 1 with default sku.

2. Managed Identities

Mandates User-Assigned Managed Identities for model interactions. No persistent connection strings or API keys.

3. Diagnostics & Logs

In-built routing profiles sending token usage, query profiles, and billing tags to centralized Log Analytics.

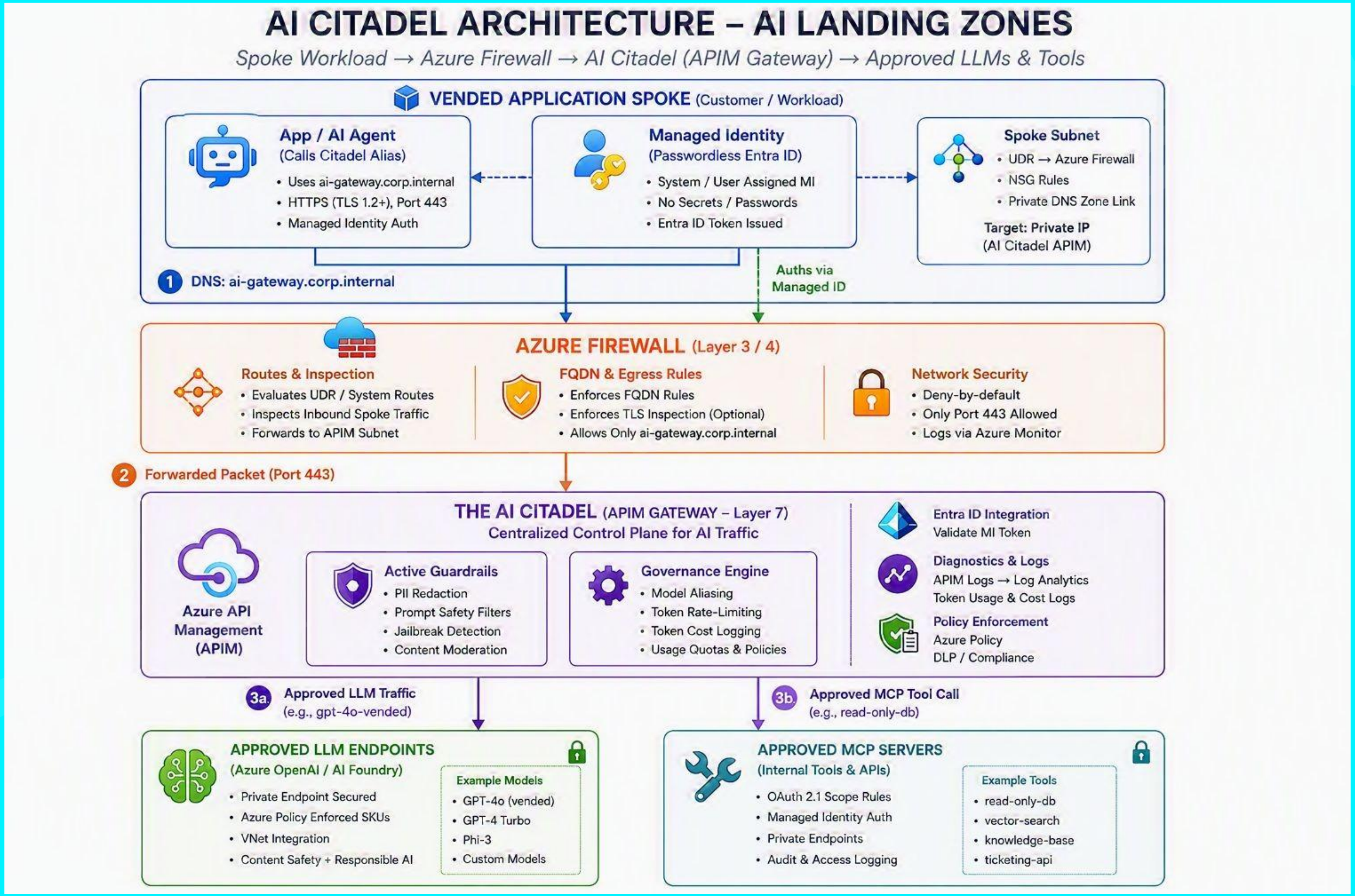
The AI Citadel Integration

Scale Meets Control

The AI Citadel design pattern balances local team speed with enterprise risk management. Local workloads run AI agents inside protected spoke environments, but they communicate outside only through verified channels.

- **Centralized Gateway:** The Spoke securely communicates back to the central Hub via dedicated API gateways.
- **Access Contract Enforcement:** Vending pipelines automatically secure and register the connection handshake.

Architectural Topology



Active Proxy vs. Passive Policy

Passive Guardrails (Azure Policy ARM)

Evaluates resources after configuration or blocks deployment outright if rules are broken.

- Blocks public endpoints.
- Cannot audit active run-time data.

Active Proxy (Traffic Inspection)

Azure API Management (APIM) intercepts data payloads dynamically in real-time.

- Strips PII / sensitive data on the fly.
- Controls costs via token rate-limiting.

Agentic Lifecycle Management

Beyond Static IaC Templates

Deploying compliance profiles at creation is stage one. Autonomous platforms require active lifecycle monitoring. We leverage lightweight AI agents embedded within platform pipelines to continuously assess runtime status.

- **Automated Self-Healing:** Automatically remediate configuration drifts.
- **Shadow AI Detection:** Spot and alert on non-compliant, ad-hoc AI resources generated outside official pipelines.

The Access Contract Engine

1. Register

The IaC vending pipeline registers the Spoke's identity inside APIM, generating individual endpoint routes automatically.

2. Bind Policies

Applies tailored compliance parameters (rate quotas, system logging rules, and region-mapping) instantly to the Spoke.

3. Finalize

Constructs and secures the end-to-end Private Link connection, completing the verified Hub-Spoke communications handshake.

Platform Maturity Model

Level 2: Automated

Subscription vending with standard IaC templates. Decentralized, fast-tracked, with basic passive controls.

Level 1: Manual

Ticket-driven provisioning. Heavy deployment friction, high risk of Shadow AI, and zero consistency.

Level 3: Citadel

Active run-time governance (APIM Proxy), self-healing infrastructures, and active Agentic drift checks.

Roadmap & Takeaways

Map Your Baseline

Define enterprise compliance guidelines before provisioning. Stop relying on manual checklists.

Inject Into Vending

Build the AI layers (Private Endpoints, Managed Identity) directly inside your Terraform/Bicep pipelines.

Go Active via APIM

Establish APIM as your inline AI Gateway proxy for real-time traffic monitoring and data governance.

Demo Time

Sort of…

Resolve issues with a disabled subscription

Your account or subscription is disabled for security reasons.

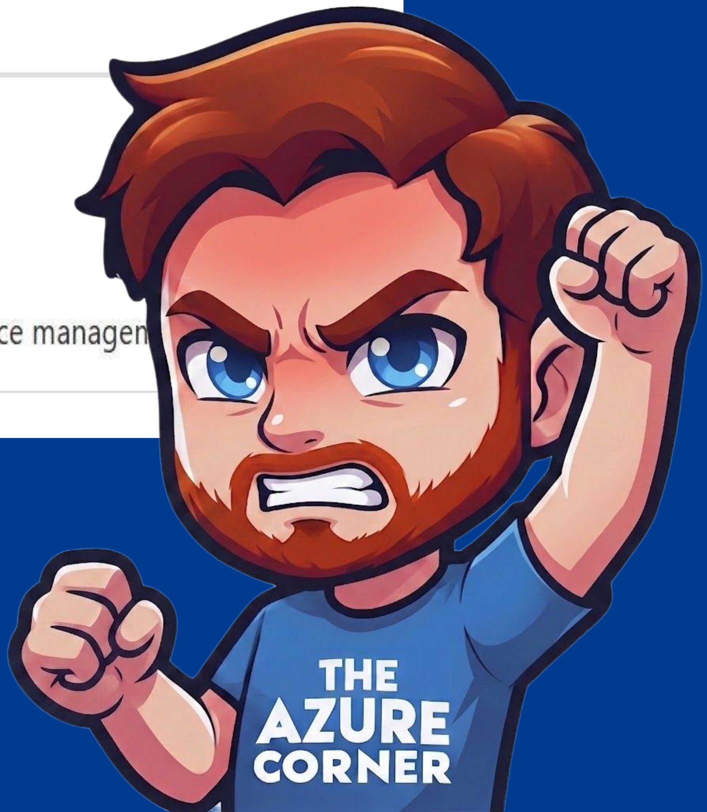
We detected suspicious activity or terms of use violations and disabled your subscription to protect the owner of the payment instrument and Microsoft. If you believe this is an error, continue to submit a support ticket with the security team to review the decision.

Relevant results from the web

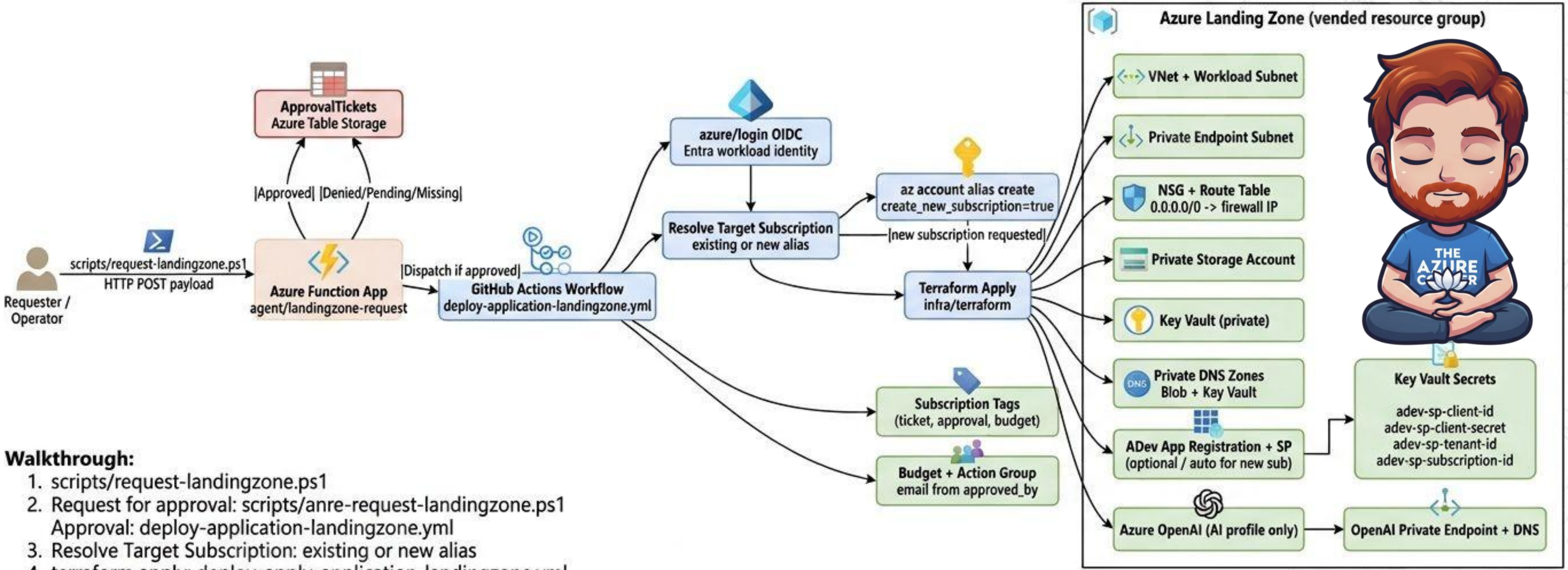
Azure Docs

Azure subscription states - Microsoft Cost Management

Learn about the different states of an Azure subscription, including active, deleted, and disabled states, and how they affect resource managem



End-to-End Architecture Visual Walkthrough



Walkthrough:

1. scripts/request-landingzone.ps1
2. Request for approval: scripts/anre-request-landingzone.ps1
   Approval: deploy-application-landingzone.yml
3. Resolve Target Subscription: existing or new alias
4. terraform apply: deploy-apply-application-landingzone.yml
5. GitHub Actions Workflow | approved

Visio notes:

- scripts/request-landingzone.ps1 attribute create-azure-application-landingzone.yml
- Prepararation/request-landingzone.ps1: HTTP POST payload
- Workflow description uses the control structure Function App the agent/landingzone-request
- Budget + Action Group email confimation landingzone.yml

Visio notes:

- Draw Azure Landing Zone (vended resource group)
  - Azure subscription management (ticket, approval, budget)
  - Management group structure (vended resource group)
- Private DNS Zones link in to all VNETs (vended resource group)
- Subscription Tags – NSG is automatically configured
- Provisioning process (vended resource group)

Questions?

Thank you for attending this MVP session!

Curtis Milne | Microsoft MVP